

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12388273
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Fedigan; Stephen John et al.

Power management unit architecture

Abstract

A power management unit architecture including a set of batteries connected to a load via a bus; a controller in operative communication with the batteries; a telemetry unit in operative communication with the controller and the batteries; at least one power switch in operative communication with the controller, the batteries and the bus; a squib unit operatively connected to the batteries; wherein the controller is configured to pull a battery current from a first battery and drive the first battery current into a second battery responsive to a predetermined current.

Inventors: Fedigan; Stephen John (Marana, AZ), Velagapudi; Mahitha (Allen, TX), Sweer; Grant Allen (Owens Cross Roads, AL), Budisalich; Derek L. (Decatur, AL), Maugans; Eric C. (Tucson, AZ), Thompson; Andrew (Vail, AZ)

Applicant: Raytheon Company (Waltham, MA)

Family ID: 90364328

Assignee: Raytheon Company (Arlington, VA)

Appl. No.: 18/131171

Filed: April 05, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240339847 A1	Oct. 10, 2024

Publication Classification

Int. Cl.: H02J7/00 (20060101); H01M6/36 (20060101); H01M6/50 (20060101)

U.S. Cl.:

CPC **H02J7/0025** (20200101); **H01M6/36** (20130101); **H01M6/5011** (20130101);
H01M6/5033 (20130101); **H01M6/5038** (20130101); **H01M6/5044** (20130101);
H02J7/0047 (20130101);

Field of Classification Search

CPC: H02J (7/0025); H02J (7/0047); H02J (7/342); H01M (6/36); H01M (6/5011); H01M (6/5033); H01M (6/5038); H01M (6/5044)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3639773	12/1971	Ayd, III	307/66	H01M 6/5011
3693068	12/1971	Bogue	320/124	H01M 6/50
3767933	12/1972	Bogue	307/66	H02J 9/06
7893561	12/2010	Weidenheimer	307/71	H02J 7/0024
8330419	12/2011	Kim	320/120	H01M 10/4257
8508191	12/2012	Kim	320/132	H02J 7/0024
10529995	12/2019	Miller, Jr.	N/A	H02J 7/0045
2022/0220867	12/2021	O'Donnell	N/A	F03G 6/071

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
60124188	12/2006	DE	F42C 15/40

OTHER PUBLICATIONS

International Search Report and Written Opinion of the International Searching Authority for Application No. PCT/US2024/014208 dated Jun. 6, 2024, 10 pp. cited by applicant
Dagarin et al., Galileo Probe Battery System—an Update, Hughes Space and Communications Company, Battery Conference on Applications and Advances, Eleventh Annual, Long Beach, CA, IEEE, Jan. 1996, 7 pp. cited by applicant

Primary Examiner: Fureman; Jared
Assistant Examiner: Warmflash; Michael J
Attorney, Agent or Firm: Bachman & LaPointe, P.C.

Background/Summary

BACKGROUND

- (1) The present disclosure is directed to a power management unit architecture that enables sequential firing of batteries.
- (2) Certain critical power applications require an electrical power source capable of ultrahigh reliability and ultralow maintenance and virtually unlimited shelf life. In such batteries, the electrodes are fully assembled for operation, but the electrolyte is held in reserve in a separate

container which may be within the battery container. Since there is no consumption of the electrodes under these circumstances, the shelf life of the battery is essentially indefinite. However, once the electrolyte is released from its reserve container, such as by mechanical puncture, explosive squib rupture or by any other means as are well known in the art, the battery is activated and thereafter has a limited standby life. Thermal batteries are useful for applications requiring extended storage time because they avoid deterioration of the active materials during storage and eliminate the loss of capacity due to self-discharge. A key feature is that the electrolyte is frozen at room temperature and is melted by the activation of heat pellets. Thermal batteries can have multiple chemistries. For example, a eutectic mixture of inorganic salts with inorganic binder can serve as the electrolyte between the anode and the cathode. A conductive heat source, consisting of iron and potassium perchlorate, is placed between each cell. When initiated, the heat pellets ignite, releasing heat and melting the eutectic electrolyte, producing voltage and current.

(3) A thermal battery is totally inert and non-reactive until activated. Because most external environments have little or no effect on the inactivated battery, it can be stored for 20+ years. The battery can be activated at any time without preparation and will begin supplying power almost immediately. After activation, the battery quickly reaches peak voltage, which declines gradually during the rest of its active life. Once activated, the battery functions until a critical active material is exhausted or until the battery cools below the electrolyte's melting point.

(4) Thermal batteries can be utilized in groups such that a first battery can be activated and at a future point in time, a subsequent battery can be activated to take on the load. However, the first battery being taken offline is susceptible to thermal runaway conditions if the power being produced is not properly managed. The offline battery can go into a thermal runaway condition and have venting problems unless a minimum current is drawn from the offline battery to a point in time when the voltage drops below a value of 10% of full voltage. Typically, power resistors have been used to dissipate the offline thermal battery power being produced. However, the heat dissipated in the power resistors can create issues with surrounding electronics in the battery staging electronics module causing premature failure. The size of the power resistors can present electronics packaging issues as well.

(5) What is needed is a power management unit architecture that allows for sequential firing of thermal batteries extending available battery power as well as allowing for offline battery management without the unwanted heat loads in the battery staging electronics package.

SUMMARY

(6) In accordance with the present disclosure, there is provided a power management unit architecture comprising a first battery in operative communication with an electrical load via a bus; a second battery in operative communication with the electrical load via the bus and in operative communication with the first battery; an nth battery in operative communication with electrical load via the bus and in operative communication with an n-1th battery; a controller in operative communication with the first battery, the second battery and up to the nth battery; a first telemetry unit in operative communication with the controller and the first battery; a second telemetry unit in operative communication with the controller and the second battery; an nth telemetry unit in operative communication with the controller and the nth battery; a first power switch in operative communication with the controller, the first battery and the bus; a second power switch in operative communication with the controller, and the second battery and the bus; an nth power switch in operative communication with the controller, and the nth battery and the bus; a first squib unit operatively connected to the first battery; a second squib unit operatively connected to the second battery; an nth squib unit operatively connected to the nth battery, wherein the controller is configured to pull a first battery current from the first battery and drive the first battery current into the second battery responsive to a predetermined current; and the controller is configured to pull a second battery current from the second battery and drive the second battery current to an n-1th battery up to the nth battery responsive to a predetermined current.

(7) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the first battery is a thermal battery; the second battery is a thermal battery; and up to the nth battery is a thermal battery.

(8) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the first battery is activated and connected to the electrical load before the second battery is activated and connected to the electrical load and an n-1th battery is activated and connected to the electrical load before the nth battery is activated and connected to the electrical load.

(9) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the controller is configured to prevent the offline first battery from overheating and out-gassing and prevent the offline second battery from overheating and out-gassing.

(10) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include periodically, at least one of a first battery current, a second battery current and up to an nth battery current are utilized in at least one of being applied to the load, charging another battery, being dumped as thermal energy and combinations thereof.

(11) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the controller is configured to pull at least one of the first battery current from the first battery and the second battery current from the second battery and up to an n-1th battery until a battery voltage reaches a target voltage of 10 percent of a full voltage value.

(12) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the controller is configured to transfer a charge from the first battery to the second battery, wherein the first battery is offline and the second battery is online; and wherein the controller is configured to transfer a charge from an n-1th battery to the nth battery, wherein the n-1th battery is offline and the nth battery is online.

(13) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the first telemetry unit, second telemetry unit and up to nth telemetry unit are configured to determine at least one of a battery impedance, a battery voltage, and a battery temperature.

(14) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the power management unit architecture further comprising a first squib multiplexer in operative communication with the first squib unit; a second squib multiplexer in operative communication with the second squib unit; and up to an nth squib multiplexer in operative communication with the nth squib unit; a squib driver circuit unit in operative communication with each of the first squib unit, the second squib unit and up to the nth squib unit; a firing interlock logic unit in operative communication with each of the first squib unit, the second squib unit and up to the nth squib unit, the controller and a guidance electronics unit configured to activate each of the first battery, the second battery and up to the nth battery.

(15) In accordance with the present disclosure, there is provided a process for sequential activation of batteries connected to an electrical load comprising operatively connecting a first battery with the electrical load; subsequently operatively connecting a second battery with the electrical load and operatively connecting the second battery in communication with the first battery; operatively connecting a controller in operative communication with the first battery and the second battery; operatively connecting a first telemetry unit in operative communication with the controller and the first battery; operatively connecting a second telemetry unit in operative communication with the controller and the second battery; and configuring the controller to pull a first battery current from the first battery and drive the first battery current into the second battery responsive to a predetermined current.

(16) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the first battery is a thermal battery and the second battery is a thermal battery.

- (17) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the process further comprising activating and connecting the first battery to the electrical load before activating and connecting the second battery to the electrical load.
- (18) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the process further comprising configuring the controller to pull the first battery current from the first battery until the first battery voltage reaches a target voltage of 10 percent of a full voltage value.
- (19) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the process further comprising preventing an offline first battery from overheating and out-gassing and preventing an offline second battery from overheating and out-gassing.
- (20) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the process further comprising utilizing at least one of a first battery current, and a second battery current in at least one of connecting to the load, charging another battery, dumping thermal energy and combinations thereof.
- (21) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the process further comprising configuring the controller to transfer a charge from the first battery to the second battery responsive to the first battery being offline and the second battery being online.
- (22) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the process further comprising determining at least one of a battery impedance, a battery voltage, and a battery temperature by using at least one of the first telemetry unit, and the second telemetry unit.
- (23) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the process further comprising operatively connecting a first squib multiplexer in with a first squib unit operatively connected to the first battery; operatively connecting a second squib multiplexer with the second squib unit operatively connected to the second battery; operatively connecting a squib driver circuit unit with each of the first squib unit and second squib unit; and operatively connecting a firing interlock logic unit with each of the first squib unit and second squib unit; and configuring the controller and a guidance electronics unit to activate each of the first battery and second battery.
- (24) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the process further comprising controlling a first power switch with the controller; controlling a second power switch with the controller.
- (25) Other details of the power management unit architecture are set forth in the following detailed description and the accompanying drawing wherein like reference numerals depict like elements.
-

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic representation of an exemplary power management unit architecture.

DETAILED DESCRIPTION

(2) Referring now to FIG. 1, a power management unit architecture **10** is shown. A grouping of batteries **12** can be used to energize an electrical load **14**. The electrical load **14** can be part of a missile, such as a control actuation system (not shown). The batteries **12** can be thermal batteries, lead acid batteries, Li-Ion and the like. The batteries **12** are connected to a bus **16** with a positive side **18** and a negative side **20**. A first battery **22** is activated and connected to the electrical load **14** before a second battery **24** is activated and connected to the electrical load **14**. Lastly, a third battery or nth battery **26** can be activated and connected to the electrical load **14** subsequently after

the second battery **24**. It is contemplated that any number of batteries **12** can be utilized depending on the mission and needs of the load **14**. The bus **16** connects the batteries **12** to the load **14**. The electrical load **14** can be supplied sequentially by a first battery current **28** from the first battery **22**, and a second battery current **30** from the second battery **24**, and an nth battery current **32** from the nth battery **26**.

(3) The power management unit architecture **10** includes the first battery **22** in operative communication with the electrical load **14**. The second battery **24** can be in operative communication with the electrical load **14** and in operative communication with the first battery **22**. The nth battery **26** can be in operative communication with the electrical load **14** and in operative communication with the first battery **22**, the second battery **24** and up to an n-1th battery.

(4) A digital signal processing system or simply controller **34** can be in operative communication with the first battery **22**, the second battery **24** and all batteries up to the nth battery **26**. The controller **34** can include hardware, firmware, and/or software components that are configured to perform the functions disclosed herein, including the functions of the power management unit architecture **10**. While not specifically shown, the controller **34** may include other computing devices (e.g., servers, mobile computing devices, FPGAs, programmable logic devices, etc.) which may be in communication with each other and/or the controller **34** via a communication network **36** to perform one or more of the disclosed functions. The controller **34** may include at least one processor **38** (e.g., a controller, microprocessor, microcontroller, digital signal processor, etc.), memory **40**, and an input/output (I/O) subsystem **42**. The controller **34** may be embodied as any type of computing device e.g., a network of computers, a combination of computers and other electronic devices, or other electronic devices. Although not specifically shown, the I/O subsystem **42** typically includes, for example, an I/O controller, a memory controller, and one or more I/O ports. The processor **38** and the I/O subsystem **42** are communicatively coupled to the memory **40**. The memory **40** may be embodied as any type of computer memory device (e.g., volatile memory such as various forms of random access memory).

(5) The first battery **22** is the initial battery to be activated. At a predetermined state, the second battery **24** can be activated and up to the nth battery **26** can be activated in sequence. For a very brief period of time, both the first battery **22** and the second battery **24** are active and online with the load **14**. The second battery **24** upon activation, goes online and picks up the load duty. The first battery **22** becomes offline. The power management unit architecture **10** is configured to prevent the offline first battery **22** from overheating and out-gassing during its offline stage, since the first battery **22** is still activated and has the potential to overheat. Periodically, the first battery current **28** is drawn off the first battery **22** and utilized either in the second battery **24** or on the load **14** or dumped as thermal energy or all of the above.

(6) A first squib unit **44** is connected to the first battery. The first squib unit **44** is utilized to activate the first battery **22**. A second squib unit **46** is connected to the second battery **24**. An nth squib unit **48** is connected to the nth battery **26**. In an exemplary embodiment, the squib unit **44**, **46**, **48** is a chemical that heats an electrolyte, producing voltage and current. In an exemplary embodiment, the squib unit **44**, **46**, **48** can include a bridge wire heating element that activates heat pellets, melts electrolyte which then produces voltage and current.

(7) A first squib multiplexer (MUX) **50** is operatively connected to the first squib unit **44**. A second squib multiplexer (MUX) **52** is operatively connected to the second squib unit **46**. An nth squib multiplexer (MUX) **54** is operatively connected to the nth squib unit **48**.

(8) A squib driver circuit unit **56** is in operative communication with each of the squib units **44**, **46**, **48** via squib driver circuit **58**. The squib driver circuit unit **56** is in operative communication with the controller **34**. A squib power source **60** is operatively connected to the squib driver circuit unit **56** via a bus voltage sensor **62**. The bus voltage sensor **62** is also in operative communication with the controller **34** and a DC-DC converter **66**. The DC-DC converter **66** is connected to the bus **16**. The squib driver circuit unit **56** and DC-DC converter **66** are in operative communication with

ground **68**.

(9) A firing interlock logic unit **70** is operatively connected to the squib MUX **50, 52, 54** via a squib MUX control circuit **72**. The firing interlock logic unit **70** is operatively connected to the controller **34** via fire command circuit **74** and bus sequence circuit **76**. The firing interlock logic unit **70** can receive command inputs, such as pre-arm command, arm command and fire command from the guidance electronic unit (GEU) **78**.

(10) A first power switch **80** is in operative connection with the first battery **22**, controller **34** and bus **16**. A second power switch **82** is in operative connection with the second battery **24**, controller **34** and bus **16**. An nth power switch **84** is in operative connection with the nth battery, controller **34** and bus **16**.

(11) The power management unit architecture **10** includes a voltage sensing circuit **86** operatively connected between the controller **34** and the batteries **12**. The voltage sensing circuit **86** can be operatively connected to a first voltage sensor **88** associated with the first battery **22**. The voltage sensing circuit **86** can be operatively connected to a second voltage sensor **90** associated with the second battery **24**. The voltage sensing circuit **86** can be operatively connected to an nth voltage sensor **92** associated with the nth battery **26**. In an exemplary embodiment, the voltage sensors **88, 90, 92** can include a first resistor in series with a second resistor (not shown). The voltage sensing circuit **86** indicates a battery voltage **94**.

(12) The power management unit architecture **10** includes a temperature sensing circuit operatively connected between the controller **34** and the batteries **12**. The temperature sensing circuit **96** can be operatively connected to a first temperature sensor **98** associated with the first battery **22**. The temperature sensing circuit **96** can be operatively connected to a second temperature sensor **100** associated with the second battery **24**. The temperature sensing circuit **96** can be operatively connected to an nth temperature sensor **102** associated with the nth battery **26**.

(13) A first telemetry unit **104** can be operatively connected to the first voltage sensor **88** and the first temperature sensor **98** and operatively connected to the first battery **22** and controller **34**.

(14) A second telemetry unit **106** can be operatively connected to the second voltage sensor **90** and the second temperature sensor **100** and operatively connected to the second battery **22** and controller **34**.

(15) An nth telemetry unit **108** can be operatively connected to the nth voltage sensor **92** and the nth temperature sensor **102** and operatively connected to the nth battery **22** and controller **34**. The telemetry units **104, 106, 108** are configured for telemetry. Telemetry is an automated process used to collect measurements and other types of data remotely. The data is sent between devices and monitored for analysis to improve device performance.

(16) A current sensing circuit **110** can be operatively connected between the controller **34** and a current sensor **112** in operative connection with the positive leg **18** of the bus **16**. The current sensing circuit **52** can include a shunt circuit, a Hall circuit, a current transformer, and the like. The current sensing circuit **52** indicates the current between the bus **16** and the batteries **12**.

(17) The power switches **80, 82, 84** can enable a current path between an offline battery **22** and an online battery **24**. The power switches **80, 82, 84** can be configured with a transistor, such as for example a power transistor, a Field Effect Transistor (FET), a Metal Oxide Semiconductor Field Effect Transistor (MOSFET), insulated-gate bipolar transistor (IGBT) and the like. The power switches **80, 82, 84** can be configured with a diode, such as a body diode. The power switches **80, 82, 84** can be coupled on the low side of the load **14**. The power switches **80, 82, 84** can have the transistor connected to a ground (not shown). The power switches **80, 82, 84** can be in operative communication with an inductor (not shown). The controller **34** can operate the power switches **80, 82, 84** to alter the pathways for the current **28, 30, 32** to flow.

(18) The power switches **80, 82, 84** can utilize silicon carbide technology to improve the packaging and the power density. The power switches **80, 82, 84** can be configured to be scaled. The power switches **80, 82, 84** can enable a path for reverse current into one of the batteries **12** during

regeneration from the control actuation system (not shown).

(19) The controller **34** can include an inner loop **114**, the inner loop **114** can be configured to control the first battery current **28**. The inner loop **114** can be configured to measure a loop error and apply one of a full voltage or no voltage for a predetermined period of time.

(20) The controller **34** can include an outer loop **116**. The outer loop **116** can be configured to control the battery **12** voltage.

(21) The controller **34** is configured to transfer a charge from the first battery **22** to the second battery **24** when the first battery **22** is offline and the second battery **24** is online. The controller **34** is configured to measure a battery impedance. The controller **34** is configured to determine an online battery resistance responsive to each of a measurement of the online battery voltage when an inductor (not shown) is connected to the online battery and measurement of the online battery voltage when the inductor is disconnected from the online battery.

(22) The power management unit architecture begins operation when the firing interlock logic **70** receives a battery firing pulse or firing command sequence from the GEU **78**. Once the first battery **22** is fired, the on-board controller **34** collects voltage, current and temperature telemetry. Based on this data, the controller **34** assesses the state-of-charge (SOC) and impedance of the battery **22**. If the battery **22** falls below a desired SOC or the battery **22** cannot deliver sufficient power to meet demands, the controller **34** automatically fires the next battery **24**, using the operating battery **22** as a power source and an on-board squib driver **56**. To conserve board real-estate, the squib driver **56** can be connected to any of the battery bridge wires thru a bank of load switches. After the battery **24** comes up to its desired voltage, the discharged battery **22** is taken offline and the new battery **24** is connected to the bus **16** thru a power switch topology. The offline battery **22** can be connected to a dissipation resistor to prevent thermal runaway. This process continues and ultimately employs the nth battery until the mission is completed.

(23) A technical advantage of the disclosed power management unit architecture includes enabling thermal batteries to support mission durations greater than 10 minutes.

(24) Another technical advantage of the disclosed power management unit architecture includes better battery utilization, significantly reduces heat generation.

(25) Another technical advantage of the disclosed power management unit architecture includes smart battery firing decisions made by the controller based on collected battery voltage, current, and temperature telemetry.

(26) Another technical advantage of the disclosed power management unit architecture includes squib power for subsequent firings provided by either external squib power source or an active battery thru a buck regulator.

(27) Another technical advantage of the disclosed power management unit architecture includes a system that has a common squib driver multiplexed across all battery stages using HS/LS power switches. HS/LS power switches isolate bridge wires in the event of a case short.

(28) Another technical advantage of the disclosed power management unit architecture includes batteries are taken on and offline by SiC switch blocks.

(29) Another technical advantage of the disclosed power management unit architecture includes switch blocks that can act as synchronous rectifiers, control dumping and enable regeneration.

(30) Another technical advantage of the disclosed power management unit architecture includes and interlock logic that inhibits the controller from firing any subsequent batteries until the source fire command is received from the guidance electronic unit (GEU).

(31) Another technical advantage of the disclosed power management unit architecture includes an independent logic power supply which enhances safety by permitting the squib power supply to be switched upstream.

(32) Another technical advantage of the disclosed power management unit architecture includes an electronic switch with resistor which enables load impedance measurements and Electrochemical Impedance Spectroscopy (EIS).

(33) Another technical advantage of the disclosed power management unit architecture includes an initial guidance electronics unit firing can be performed by either directly or through a digital interface.

(34) Another technical advantage of the disclosed power management unit architecture includes a logic that allows for the first battery to be fired at the beginning of the mission, and subsequent battery stages can be fired by the controller based on the battery capability to support longer missions.

(35) Another technical advantage of the disclosed power management unit architecture includes a flexible architecture scalable to higher or lower power systems.

(36) Another technical advantage of the disclosed power management unit architecture includes battery telemetry information which permits the controller to make smart firing decisions, based on the online battery's impedance or state-of-charge.

(37) Another technical advantage of the disclosed power management unit architecture includes telemetry which includes battery voltage, bus current, and environmental temperature.

(38) There has been provided a power management unit architecture. While the power management unit architecture has been described in the context of specific embodiments thereof, other unforeseen alternatives, modifications, and variations may become apparent to those skilled in the art having read the foregoing description. Accordingly, it is intended to embrace those alternatives, modifications, and variations which fall within the broad scope of the appended claims.

Claims

1. A power management unit architecture comprising: a first battery in operative communication with an electrical load via a bus; a second battery in operative communication with the electrical load via the bus and in operative communication with the first battery; an nth battery in operative communication with electrical load via the bus and in operative communication with an n-1th battery; a controller in operative communication with the first battery, the second battery and up to the nth battery; a first telemetry unit in operative communication with the controller and the first battery; a second telemetry unit in operative communication with the controller and the second battery; an nth telemetry unit in operative communication with the controller and the nth battery; a first power switch in operative communication with the controller, the first battery and the bus; a second power switch in operative communication with the controller, and the second battery and the bus; an nth power switch in operative communication with the controller, and the nth battery and the bus; a first squib unit operatively connected to the first battery; a second squib unit operatively connected to the second battery; an nth squib unit operatively connected to the nth battery, wherein the controller is configured to pull a first battery current from the first battery and drive the first battery current into the second battery responsive to a predetermined current; and the controller is configured to pull a second battery current from the second battery and drive the second battery current to an n-1th battery up to the nth battery responsive to a predetermined current.
2. The power management unit architecture according to claim 1, wherein the first battery is a thermal battery; the second battery is a thermal battery; and up to the nth battery is a thermal battery.
3. The power management unit architecture according to claim 1, wherein the first battery is activated and connected to the electrical load before the second battery is activated and connected to the electrical load and an n-1th battery is activated and connected to the electrical load before the nth battery is activated and connected to the electrical load.
4. The power management unit architecture according to claim 3, wherein the controller is configured to prevent the offline first battery from overheating and out-gassing and prevent the offline second battery from overheating and out-gassing.

5. The power management unit architecture according to claim 4, wherein periodically, at least one of a first battery current, a second battery current and up to an nth battery current are utilized in at least one of being applied to the load, charging another battery, being dumped as thermal energy and combinations thereof.
6. The power management unit architecture according to claim 1, wherein the controller is configured to pull at least one of the first battery current from the first battery and the second battery current from the second battery and up to an n-1th battery until a battery voltage reaches a target voltage of 10 percent of a full voltage value.
7. The power management unit architecture according to claim 1, wherein the controller is configured to transfer a charge from the first battery to the second battery, wherein the first battery is offline and the second battery is online; and wherein the controller is configured to transfer a charge from an n-1th battery to the nth battery, wherein the n-1th battery is offline and the nth battery is online.
8. The power management unit architecture according to claim 1, wherein the first telemetry unit, second telemetry unit and up to nth telemetry unit are configured to determine at least one of a battery impedance, a battery voltage, and a battery temperature.
9. The power management unit architecture according to claim 1, further comprising: a first squib multiplexer in operative communication with the first squib unit; a second squib multiplexer in operative communication with the second squib unit; and up to an nth squib multiplexer in operative communication with the nth squib unit; a squib driver circuit unit in operative communication with each of the first squib unit, the second squib unit and up to the nth squib unit; a firing interlock logic unit in operative communication with each of the first squib unit, the second squib unit and up to the nth squib unit, the controller and a guidance electronics unit configured to activate each of the first battery, the second battery and up to the nth battery.
10. A process for sequential activation of batteries connected to an electrical load comprising: operatively connecting a first battery with the electrical load; subsequently operatively connecting a second battery with the electrical load and operatively connecting the second battery in communication with the first battery; operatively connecting a controller in operative communication with first the battery and the second battery; operatively connecting a first telemetry unit in operative communication with the controller and the first battery; operatively connecting a second telemetry unit in operative communication with the controller and the second battery; and configuring the controller to pull a first battery current from the first battery and drive the first battery current into the second battery responsive to a predetermined current.
11. The process of claim 10, wherein the first battery is a thermal battery and the second battery is a thermal battery.
12. The process of claim 10, further comprising: activating and connecting the first battery to the electrical load before activating and connecting the second battery to the electrical load.
13. The process of claim 10, further comprising: configuring the controller to pull the first battery current from the first battery until the first battery voltage reaches a target voltage of 10 percent of a full voltage value.
14. The process of claim 13, further comprising: preventing an offline first battery from overheating and out-gassing and preventing an offline second battery from overheating and out-gassing.
15. The process of claim 10, further comprising: utilizing at least one of a first battery current, and a second battery current in at least one of connecting to the load, charging another battery, dumping thermal energy and combinations thereof.
16. The process of claim 10, further comprising: configuring the controller to transfer a charge from the first battery to the second battery responsive to the first battery being offline and the second battery being online.
17. The process of claim 10, further comprising: determining at least one of a battery impedance, a

battery voltage, and a battery temperature by using at least one of the first telemetry unit, and the second telemetry unit.

18. The process of claim 10, further comprising: operatively connecting a first squib multiplexer in with a first squib unit operatively connected to the first battery; operatively connecting a second squib multiplexer with the second squib unit operatively connected to the second battery; operatively connecting a squib driver circuit unit with each of the first squib unit and second squib unit; and operatively connecting a firing interlock logic unit with each of the first squib unit and second squib unit; and configuring the controller and a guidance electronics unit to activate each of the first battery and second battery.

19. The process of claim 10 further comprising: controlling a first power switch with the controller; controlling a second power switch with the controller.
